

Introduction to Information Retrieval

Module 1: Basic Concepts & The Retrieval Process

Reference: Manning, Raghavan, & Schütze (Cambridge University Press)

The Motivation

The Information Overload

We are living in an era of unstructured data. From 100 trillion web pages to millions of internal corporate documents, the challenge isn't finding data, but finding **relevant** information.

IR allows us to satisfy an "Information Need" in a world where browsing is no longer sufficient.



Basic Concepts



Document

The basic unit we retrieve (e.g., a web page, a PDF, or a paragraph).



Query

The user's expression of their information need (usually keywords).



Relevance

How well the retrieved document satisfies the user's specific need.

IR vs. Data Retrieval

Feature	Data Retrieval (DB)	Information Retrieval (IR)
Matching	Exact Match (SQL)	Partial/Approximate Match
Model	Deterministic	Probabilistic
Results	Unordered Set	Ranked List by Relevance
Inference	Deductive	Inductive

The Retrieval Process

User Side

1. Identify Problem
2. Formulate Information Need
3. Generate Query
4. Evaluate Results (Feedback)

System Side

1. Document Collection
2. Indexing (Preprocessing)
3. Search & Matching
4. Ranking Documents

Step-by-Step Logic

Crawler

Gathering Docs

Query

User Input

Ranker

Sorted Results

Indexer

Building the Index

Matcher

Calculating Scores



Component: Indexing

Why Index?

Linear scanning of 100TB of data is impossible for real-time search. We need an **Inverted Index**.

- **Tokenization:** Breaking text into terms.
- **Normalization:** Handling case/punctuation.
- **Postings List:** Mapping terms to DocIDs.

 Indexing Server Room

Component: Query Processor



The Bridge to Knowledge

The Query Processor interprets user intent. It doesn't just look for words; it looks for meaning.

Key Tasks: Spelling correction, expansion (synonyms), and weighting terms based on importance (TF-IDF).

Parts of the System



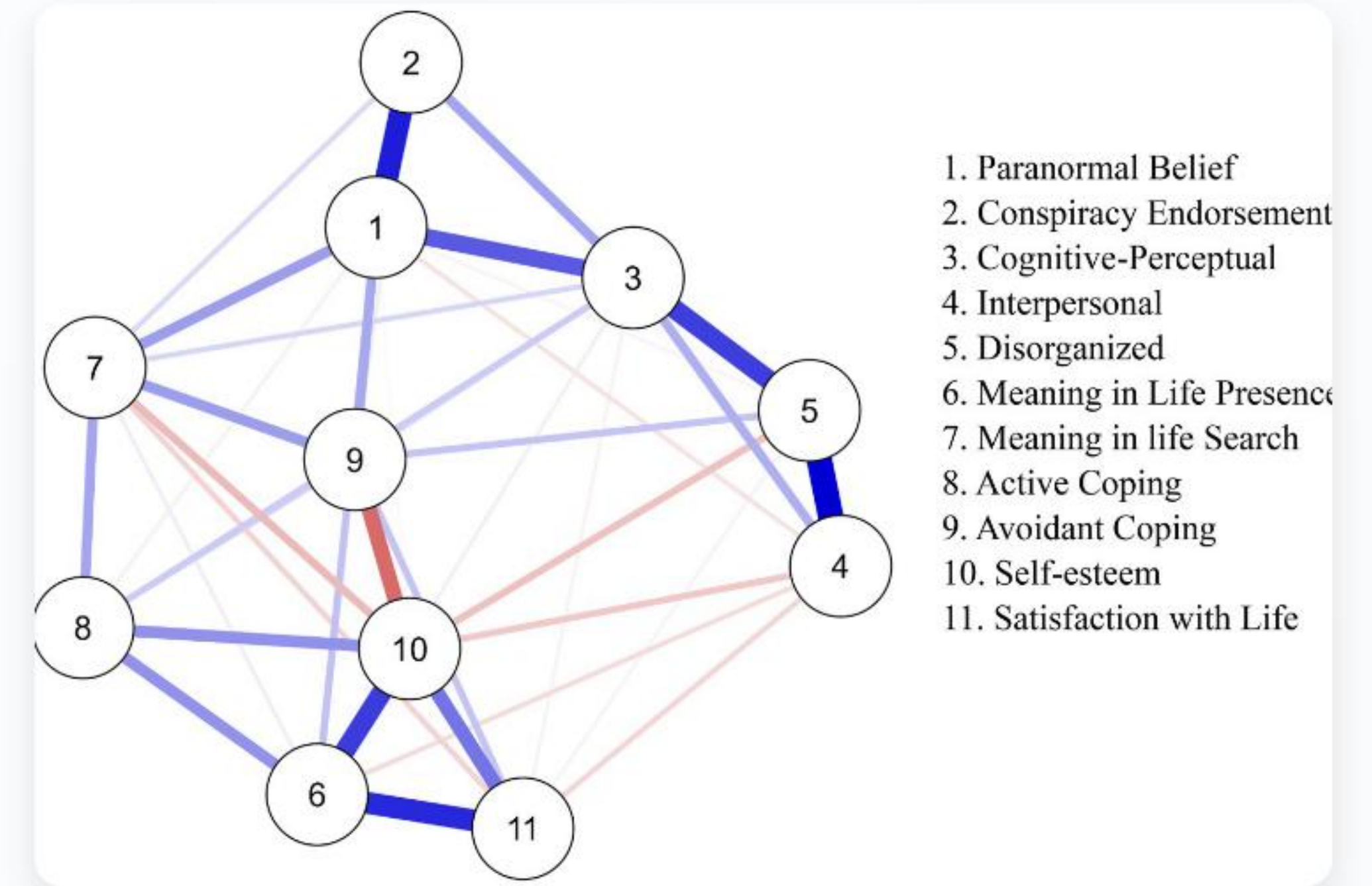
People

Users with specific information needs and diverse contexts.



Documents

The carriers of information in various formats (Text, Media).



Topics

The abstract representations and labels that link needs to docs.

| Types of Information Systems



Web IR

Scale: Billions of docs. Challenges:
Spam, hyper-linking, scale.



Enterprise

Scale: Smaller. Challenge: Security,
proprietary formats.



Desktop

Focus: Personal files. Goal: Extreme
speed and low resource use.

Measuring Success

95%

Precision vs. Recall

Quality Matters

In IR, we don't just want results; we want the **right** results.

Precision: Percentage of retrieved docs that are relevant.

Recall: Percentage of all relevant docs that were retrieved.